WORKED SOLUTION

STEP II 2014, Question 6

Pure • Trigonometric Series

STEP II 2014, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q6

The key idea

Multiplying each $\cos rx$ by $2\sin\frac{1}{2}x$ turns it into a difference of two sines, so the sum telescopes. That sum is exactly $S_n'(x)$, which is why the identity is there: it tells us where S_n is stationary. At such a point the new term $\frac{1}{n}\sin nx$ turns out to be non-negative, so adding it cannot create a negative minimum, and an induction starting from $S_1 = \sin x$ finishes the job.

Solution

The cosine sum

By the factor formula $\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$ with $A = (r+\frac{1}{2})x$, $B = (r-\frac{1}{2})x$,

$$\sin(r+\tfrac{1}{2})x - \sin(r-\tfrac{1}{2})x = 2\cos rx\,\sin\tfrac{1}{2}x.$$

Summing from $r = 1$ to n, the right-hand sides add to $2\sin\frac{1}{2}x\,(\cos x + \cdots + \cos nx)$, while on the left everything cancels except the first and last terms:

$$2\sin\tfrac{1}{2}x\sum_{r=1}^{n}\cos rx = \sin(n+\tfrac{1}{2})x - \sin\tfrac{1}{2}x.$$

Dividing by $2\sin\frac{1}{2}x \neq 0$ gives the required result. Note that differentiating term by term,

$$S_n'(x) = \sum_{r=1}^{n}\cos rx,$$

so the identity is a closed form for S_n'.

Part (i)

Here $S_2(x) = \sin x + \frac{1}{2}\sin 2x$, so

$$S_2'(x) = \cos x + \cos 2x = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1).$$

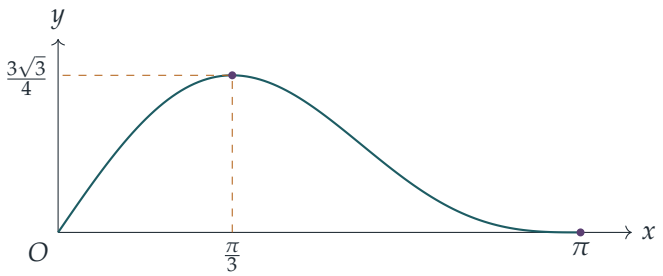
On $[0, \pi]$ this vanishes when $\cos x = \frac{1}{2}$, i.e. $x = \frac{\pi}{3}$, and when $\cos x = -1$, i.e. $x = \pi$. The values are

$$S_2(\tfrac{\pi}{3}) = \tfrac{\sqrt{3}}{2} + \tfrac{1}{2} \cdot \tfrac{\sqrt{3}}{2} = \tfrac{3\sqrt{3}}{4}, \qquad S_2(\pi) = 0.$$

Since $\cos x + 1 \geq 0$, the sign of S_2' is that of $2\cos x - 1$: positive on $[0, \frac{\pi}{3})$ and negative on $(\frac{\pi}{3}, \pi)$. So $x = \frac{\pi}{3}$ is a maximum, and S_2 decreases from there to 0 at $x = \pi$, arriving with zero gradient. Near π, putting $x = \pi - t$,

$$S_2 = \sin t - \tfrac{1}{2}\sin 2t \approx \left(t - \tfrac{t^3}{6}\right) - \left(t - \tfrac{4t^3}{6}\right) = \tfrac{1}{2}t^3,$$

so the curve flattens out like a cubic as it reaches the axis. Also $S_2(0) = 0$ with $S_2'(0) = 2$.



Part (ii)

Suppose $S_n'(x_0) = 0$ with $0 < x_0 < \pi$. Then $\sin \frac{1}{2}x_0 \neq 0$, so the identity applies and gives $\sin(n + \frac{1}{2})x_0 = \sin \frac{1}{2}x_0$. Expanding the left-hand side,

$$\sin nx_0 \cos \tfrac{1}{2}x_0 + \cos nx_0 \sin \tfrac{1}{2}x_0 = \sin \tfrac{1}{2}x_0 \iff \sin nx_0 \cos \tfrac{1}{2}x_0 = (1 - \cos nx_0)\sin \tfrac{1}{2}x_0.$$

Since $0 < \frac{1}{2}x_0 < \frac{\pi}{2}$, $\cos \frac{1}{2}x_0 > 0$, and dividing by it gives

$$\sin nx_0 = (1 - \cos nx_0)\tan \tfrac{1}{2}x_0,$$

as required. Hence, for $n \geq 2$,

$$S_n(x_0) - S_{n-1}(x_0) = \frac{1}{n}\sin nx_0 = \frac{1}{n}(1 - \cos nx_0)\tan \tfrac{1}{2}x_0 \geq 0,$$

because $1 - \cos nx_0 \geq 0$ and $\tan \frac{1}{2}x_0 > 0$. So $S_n(x_0) \geq S_{n-1}(x_0)$.

Now suppose $S_{n-1}(x) > 0$ for all $x \in (0, \pi)$, and, for a contradiction, that $S_n(c) \leq 0$ for some $c \in (0, \pi)$.

Ask yourself How do we know there is a stationary point to use?

S_n is continuous on the closed interval $[0, \pi]$, so it attains a minimum value m there, and $m \leq S_n(c) \leq 0$. If $m < 0$, the minimum is not at an end, since $S_n(0) = S_n(\pi) = 0$; if $m = 0$, then c itself is an interior point where it is attained. Either way there is

$x_0 \in (0, \pi)$ with $S_n(x_0) = m \leq 0$, and since S_n is differentiable, an interior minimum is a stationary point: $S_n'(x_0) = 0$. The previous result then gives

$$0 \geq S_n(x_0) \geq S_{n-1}(x_0) > 0,$$

a contradiction. So $S_n(x) > 0$ for all $x \in (0, \pi)$.

Part (iii)

We prove by induction that $S_n(x) > 0$ for $0 < x < \pi$. For $n = 1$, $S_1(x) = \sin x > 0$ on $(0, \pi)$. If it holds for S_{n-1}, then part (ii) shows it holds for S_n. So it holds for every $n \geq 1$. At the end points every term $\frac{1}{r}\sin rx$ is zero, so $S_n(0) = S_n(\pi) = 0$. Therefore $S_n(x) \geq 0$ for all $n \geq 1$ and $0 \leq x \leq \pi$.

Checks

- Put $n = 1$ in the cosine sum: $\dfrac{\sin \frac{3}{2}x - \sin \frac{1}{2}x}{2 \sin \frac{1}{2}x}$ should be $\cos x$, and it is, by the factor formula.
- Part (i) agrees with the identity: $S_2' = 0$ means $\sin \frac{5}{2}x = \sin \frac{1}{2}x$, whose solutions in $(0, \pi]$ are $x = \frac{\pi}{3}$ and $x = \pi$.
- The maximum $\frac{3\sqrt{3}}{4} \approx 1.299$ matches a direct evaluation of S_2 on a fine grid. A computer check found every stationary point of S_n in $(0, \pi)$ for $2 \leq n \leq 40$ and confirmed the identity of part (ii) and $S_n(x_0) \geq S_{n-1}(x_0)$ at each, and that $S_n > 0$ inside the interval for $n \leq 60$.

Common slips

- Arguing "$S_n \geq S_{n-1}$ everywhere". That is false (for example $\sin 2x < 0$ on $(\frac{\pi}{2}, \pi)$); the inequality holds only at stationary points of S_n, which is why the minimum argument is needed.
- Forgetting the end points when locating the minimum: you must use $S_n(0) = S_n(\pi) = 0$ to place a non-positive minimum inside the interval.
- Dividing by $\cos \frac{1}{2}x_0$ without saying why it is positive.

Beyond STEP

The result of part (iii) is the Fejér–Jackson–Gronwall inequality. As $n \to \infty$, $S_n(x)$ converges to $\frac{1}{2}(\pi - x)$ on $(0, \pi]$: $\sum \frac{1}{r}\sin rx$ is the Fourier series of a sawtooth wave, and near $x = 0$ the partial sums overshoot (the Gibbs phenomenon). Fourier series, and the delicate question of when they converge, are a central topic in a first analysis or methods course. T. W. Körner's *Fourier Analysis* is a wonderful, wide-ranging account.

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